

Sample Allocation and Computer-Assisted Sampling

A Technical Guidebook for Excel and Stata Implementation

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National Statistical Training Reference Guide

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Abstract

This learning guide is designed for participants of the advanced sample design session. It integrates the theoretical foundations of stratified sample allocation with clear, actionable computational strategies in Microsoft Excel and Stata. Covered methodologies include Equal, Proportional, Neyman (Optimal), Power (Square Root), and Cost-Constrained Allocation, along with practical handling of field constraints, iterative minimum sample size flooring, sample selection workflows, and design-based variance declaration.

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1 Why Allocation Matters: The Problem We Are Solving

When conducting a stratified sample survey—standard practice across modern National Statistical Offices (NSOs)—the target population is first segmented into mutually exclusive subpopulations called **strata** (such as provinces, ecological zones, urban/rural districts, or industrial sectors).

Once these strata are established, researchers face a critical statistical design choice:

Given a total sample size budget of n units, how many units should be selected from each stratum h (n_h)?

This structural distribution is called **sample allocation**. Far from a trivial administrative task, it directly influences the statistical precision, budget efficiency, and analytical validity of the entire research project.

1.1 The Fallacy of Simple Equal Division

Consider a hypothetical household survey in Nepal. The budget dictates a total sample of $n = 3,000$ households spread across the 7 provinces. A naive planner might divide the sample equally: $3,000 \div 7 \approx 429$ households per province. Table 1 illustrates the resulting statistical imbalance when contrasted with the actual population distribution.

Table 1: Nepal Household Distribution vs. Equal Sample Allocation

Province	Population (Households)	Equal Allocation (n_h)	% of Pop.	Sampling Fraction (f_h)
Koshi	720,000	429	18.5%	0.060%
Madhesh	810,000	429	20.8%	0.053%
Bagmati	960,000	429	24.6%	0.045%
Gandaki	360,000	429	9.2%	0.119%
Lumbini	630,000	429	16.2%	0.068%
Karnali	180,000	429	4.6%	0.238%
Sudurpashchim	240,000	429	6.2%	0.179%
Total	3,900,000	3,003	100.0%	0.077%

1.2 Analytical Variables Controlled by Allocation

An allocation plan determines two fundamental statistical elements:

1. **Sampling Fraction (f_h):** The proportion of the stratum population selected into the sample:

$$f_h = \frac{n_h}{N_h} \quad (1)$$

2. **Design Weights (w_h):** To correct for unequal selection probabilities and ensure unbiased national estimates, sample observations must be weighted by the inverse of their sampling fraction:

$$w_h = \frac{1}{f_h} = \frac{N_h}{n_h} \quad (2)$$

Under poor or unplanned allocations, design weights become highly volatile, which drastically inflates the overall variance (standard errors) of national estimators.

1.3 The Three Pillars of Stratified Allocation

Robust sample designs balance three competing characteristics:

- **Population Size (N_h):** Larger strata naturally require larger samples to prevent extreme differences in selection probabilities.
- **Internal Variability (S_h):** Strata containing high internal diversity (e.g., highly unequal incomes) require larger samples to achieve stable estimations. Homogeneous strata can be estimated precisely with very few units.
- **Sampling Unit Cost (c_h):** Remote mountainous geographical areas demand significantly higher travel, transport, and interview costs than dense urban centers. Wise designs shift sample sizes to cost-efficient sectors when variance targets permit.

Table 2: Sampling Vulnerabilities Caused by Poor Allocation Decisions

Design Error		Statistical or Operational Consequence
Over-sampling large strata	large	Wastes fieldwork budget; suppresses small-domain representation.
Under-sampling small strata	small	Estimates for minority groups or critical administrative divisions become unstable.
Ignoring internal variability		High-variance strata are under-sampled, causing unacceptably large standard errors for national aggregates.
Ignoring operational field costs		Triggers severe budget overruns or premature halting of fieldwork.
Unregulated rounding		Total final sample size drifts away from fixed contractual targets.
Absence of stratum-level sample floors		Some strata receive 0 or 1 unit, rendering standard error and variance estimation mathematically impossible.

Key Takeaway

Allocation is not a clerical chore; it is an active statistical decision. Every single value entered into an allocation matrix directly controls the precision, operational budget, and national validity of the final survey outputs.

2 Equal and Proportional Allocation Methods

To build a solid practical foundation, we begin with the classic approaches: Equal and Proportional Allocation. To preserve continuity, we will use a single running survey dataset throughout this guidebook.

2.1 The Running Reference Dataset

We are planning a national Household Income Survey. The country is stratified into $H = 5$ distinct socio-ecological strata. The complete fieldwork budget accommodates a total sample size of $n = 1,200$ households.

Table 3: Baseline Strata Information (Household Income Survey)

Stratum (h)	Geographical Description	Population (N_h)	Income Std. Dev. (S_h , NPR '000)
1	Mountain Rural	80,000	12.0
2	Hill Rural	340,000	15.0
3	Hill Urban	260,000	22.0
4	Terai Rural	780,000	18.0
5	Terai Urban	440,000	28.0
Total		1,900,000	

2.2 Equal Allocation

Equal allocation distributes the total sample budget n uniformly across the H strata:

$$n_h = \frac{n}{H} \quad (3)$$

Applying this to our survey: $n_h = \frac{1200}{5} = 240$ households per stratum.

Table 4: Equal Allocation Results

Stratum	Description	Sample Size (n_h)	Sampling Fraction (f_h)	Design Weight (w_h)
1	Mountain Rural	240	0.3000%	333.3
2	Hill Rural	240	0.0706%	1,416.7
3	Hill Urban	240	0.0923%	1,083.3
4	Terai Rural	240	0.0308%	3,250.0
5	Terai Urban	240	0.0545%	1,833.3
Total		1,200		

When Equal Allocation is Justified: While not optimal for nationwide estimates, equal allocation is chosen when:

- **Domain-level analysis** is the primary objective (e.g., comparing poverty metrics across regions with equal confidence, irrespective of population size).

- Strata populations are relatively uniform in size.
- Prior parameters (variances, costs) are completely unknown (e.g., pilot studies).

Warning & Field Note

Notice the extreme variance in design weights in Table 4. A household in Terai Rural carries a weight of 3,250, whereas a Mountain Rural household carries 333.3. This tenfold weight variation will drastically inflate the design effect (DEFF) and overall standard errors for national averages.

2.3 Proportional Allocation

Proportional allocation distributes units based strictly on each stratum's population share. Larger strata receive proportionally larger samples:

$$n_h = n \times \frac{N_h}{N} \quad (4)$$

where $N = \sum_{h=1}^H N_h$ represents the total population (here, 1,900,000).

Step-by-Step Manual Calculation for Stratum 4 (Terai Rural):

1. **Compute Population Weight:** $W_4 = \frac{N_4}{N} = \frac{780,000}{1,900,000} \approx 0.41053$ (or 41.05%)
2. **Apply to Total Sample:** $n_4 = 1,200 \times 0.41053 = 492.63$
3. **Round to Nearest Whole Number:** $n_4 \approx 493$

Table 5: Proportional Allocation Results

Stratum	Description	Pop. Share	Raw n_h	Rounded n_h	Sampling Fraction (f_h)
1	Mountain Rural	0.0421	50.53	51	0.0638%
2	Hill Rural	0.1789	214.74	215	0.0632%
3	Hill Urban	0.1368	164.21	164	0.0631%
4	Terai Rural	0.4105	492.63	493	0.0632%
5	Terai Urban	0.2316	277.89	277	0.0630%
Total		1.0000	1,200.00	1,200	0.0632%

Key Characteristics of Proportional Designs:

- **Constant Sampling Fraction (f_h):** Every stratum exhibits an identical sampling fraction ($\approx 0.063\%$).
- **Equal-Probability Design (EPSEM):** Every household in the population has the exact same probability of being selected.
- **Uniform Weights ($w_h \approx 1,583$):** Since weights are constant, the complex sample design collapses back to simple, self-weighting averages for aggregate estimation, minimizing weighting-related variance inflation.

2.4 Side-by-Side Comparison: Equal vs. Proportional

Comparing the two methods in Table 6 highlights the structural trade-offs.

Table 6: Equal vs. Proportional Sample Allocation Comparison

Stratum	Description	Equal n_h	Proportional n_h	Net Difference
1	Mountain Rural	240	51	−189
2	Hill Rural	240	215	−25
3	Hill Urban	240	164	−76
4	Terai Rural	240	493	+253
5	Terai Urban	240	277	+37
Total		1,200	1,200	

Equal allocation dramatically oversamples low-population areas (such as Mountain Rural, jumping from 51 to 240) at the expense of highly populated agricultural/commercial hubs (such as Terai Rural, dropping from 493 to 240).

While proportional allocation is the statistical baseline, it remains blind to internal strata variance. In our table, Terai Urban's standard deviation of income is over twice that of Mountain Rural (28.0 vs. 12.0). To estimate national income metrics with high precision, we must transition to variance-sensitive designs.

3 Neyman (Optimal) Sample Allocation

Developed by Jerzy Neyman in 1934, optimal allocation incorporates population size and internal variability simultaneously. The goal of Neyman allocation is to **minimize the variance of the overall population estimator** for a fixed total sample size n .

3.1 The Mathematical Formula

Neyman allocation distributes the sample as follows:

$$n_h = n \times \frac{N_h S_h}{\sum_{h=1}^H N_h S_h} \quad (5)$$

where:

- N_h is the population size of stratum h .
- S_h is the standard deviation of the indicator of interest within stratum h .
- $\sum N_h S_h$ is the sum of these products across all strata, acting as a normalization denominator.

3.2 Step-by-Step Allocation Workflow

Step 1: Compute Priority Products ($N_h \times S_h$) Multiply each stratum's population size by its standard deviation.

Table 7: Computing Neyman Priority Products

Stratum	Description	Population (N_h)	Std. Dev. (S_h)	Product ($N_h S_h$)
1	Mountain Rural	80,000	12.0	960,000
2	Hill Rural	340,000	15.0	5,100,000
3	Hill Urban	260,000	22.0	5,720,000
4	Terai Rural	780,000	18.0	14,040,000
5	Terai Urban	440,000	28.0	12,320,000
Sum				38,140,000

Step 2: Calculate Allocation Ratios Divide each stratum product by the overall sum (38,140,000).

- Stratum 1 ratio: $960,000/38,140,000 \approx 0.02517$
- Stratum 2 ratio: $5,100,000/38,140,000 \approx 0.13372$
- Stratum 3 ratio: $5,720,000/38,140,000 \approx 0.15000$
- Stratum 4 ratio: $14,040,000/38,140,000 \approx 0.36809$
- Stratum 5 ratio: $12,320,000/38,140,000 \approx 0.32300$

Table 8: Neyman Sample Allocations and Initial Totals

Stratum	Description	Ratio	Raw n_h	Rounded n_h
1	Mountain Rural	0.02517	30.20	30
2	Hill Rural	0.13372	160.46	161
3	Hill Urban	0.15000	180.00	180
4	Terai Rural	0.36809	441.71	442
5	Terai Urban	0.32300	387.60	388
Total		1.00000	1,200.00	1,201

Step 3: Distribute the $n = 1,200$ Sample and Round Multiply each ratio by 1,200 and round to the nearest integer.

Warning & Field Note

The rounded total is 1,201, exceeding our budget of 1,200 by one unit. This is a common rounding issue. To correct this, we subtract one unit from the stratum that was rounded up with the largest fractional remainder. Stratum 2 was rounded from 160.46 up to 161 (a drift of 0.54). Adjusting Stratum 2 down to 160 restores the total to 1,200.

3.3 Side-by-Side Comparison: Equal, Proportional, and Neyman

Table 9 displays all three classical methods side-by-side.

Table 9: Comprehensive Comparison of Classical Allocation Methods

Stratum	Description	Pop. (N_h)	SD (S_h)	Equal	Proportional	Neyman
1	Mountain Rural	80,000	12.0	240	51	30
2	Hill Rural	340,000	15.0	240	215	160
3	Hill Urban	260,000	22.0	240	164	180
4	Terai Rural	780,000	18.0	240	493	442
5	Terai Urban	440,000	28.0	240	277	388
Total		1,900,000		1,200	1,200	1,200

Analyzing these differences reveals how Neyman allocation responds to variance:

- **Terai Urban ($S_h = 28.0$):** Since income variance is high, Neyman increases its sample to 388 units (up from the proportional allocation of 277).
- **Mountain Rural ($S_h = 12.0$):** Since income variance is low (relative homogeneity), Neyman assigns it just 30 units (down from the proportional allocation of 51).
- **Hill Urban vs. Hill Rural:** Hill Urban is smaller than Hill Rural in population (260k vs. 340k) but exhibits higher internal income variance ($S_h = 22.0$ vs. 15.0). Consequently, Neyman prioritizes Hill Urban with a larger sample size (180 vs. 160).

3.4 Why Neyman is Optimally Efficient

The theoretical variance of the estimated stratified sample population mean is given by:

$$V(\bar{y}_{st}) = \sum_{h=1}^H W_h^2 \frac{S_h^2}{n_h} \quad (6)$$

Neyman allocation mathematically minimizes this expression subject to the budget constraint $\sum n_h = n$.

We define the design **efficiency gain** as:

$$\text{Efficiency Gain} = \frac{V_{\text{proportional}}(\bar{y}_{st})}{V_{\text{Neyman}}(\bar{y}_{st})} \quad (7)$$

An efficiency gain of 1.15 indicates that Neyman allocation achieves the same level of precision as a proportional sample that is 15% larger, saving substantial financial resources.

3.5 Practical Challenges of Neyman Allocation

Despite its mathematical advantages, Neyman allocation has four major practical limitations:

1. **Requirement for Prior Variances:** You must have reliable estimates of S_h from a pilot study, administrative registers, or a previous census round.
2. **Non-Uniform Weights:** Sampling fractions vary across strata. Design weights must be used during analysis to avoid biased results.
3. **Single-Indicator Optimization:** An allocation optimized for income may be highly inefficient for health, education, or labor metrics in a multi-purpose survey. In practice, researchers optimize for the primary policy indicator.
4. **Over-Allocation:** For small strata with extremely high variance, the formula can calculate $n_h > N_h$. We address this "take-all" exception in Section 7.

4 Power Allocation and Advanced Variants

The classical methods can struggle with complex real-world survey demands, such as analyzing small minority populations or managing highly unequal regional fieldwork costs. This section introduces advanced variants to handle these situations.

4.1 Power Allocation

When a stratum population is small (e.g., representing just 2% of the national total), proportional allocation assigns it too few units to support reliable regional estimates. Conversely, equal allocation oversamples small strata at the expense of national precision.

Power allocation bridges this gap using a tuning exponent q ($0 \leq q \leq 1$) to balance national and stratum-level precision:

$$n_h = n \times \frac{N_h^q}{\sum_{h=1}^H N_h^q} \quad (8)$$

The parameter q acts as a dial to adjust the allocation style:

- $q = 1.0$: Restores Proportional Allocation.
- $q = 0.0$: Restores Equal Allocation.
- $q = 0.5$: Represents **Square Root Allocation**, a common standard in international household surveys.

Step-by-Step Calculation for Square Root Allocation ($q = 0.5$):

1. Compute $\sqrt{N_h}$ for each stratum.
2. Normalize these square roots to find each stratum's allocation share.
3. Multiply by $n = 1,200$ and resolve rounding.

Table 10: Power Allocation ($q = 0.5$) Calculation Matrix

Stratum	Description	Population (N_h)	Square Root $\sqrt{N_h}$	Ratio	Rounded n_h
1	Mountain Rural	80,000	282.84	0.09679	116
2	Hill Rural	340,000	583.10	0.19952	239
3	Hill Urban	260,000	509.90	0.17449	209
4	Terai Rural	780,000	883.18	0.30222	363
5	Terai Urban	440,000	663.32	0.22699	273
Total		1,900,000	2,922.34	1.00000	1,200

Table 11 compares power allocation to the other three methods. By setting $q = 0.5$, Mountain Rural receives 116 households.

Table 11: Comparison of All Key Allocation Methods

Stratum	Description	Equal	Proportional	Neyman	Power ($q = 0.5$)
1	Mountain Rural	240	51	30	116
2	Hill Rural	240	215	160	239
3	Hill Urban	240	164	180	209
4	Terai Rural	240	493	442	363
5	Terai Urban	240	277	388	273
Total		1,200	1,200	1,200	1,200

4.2 Cost-Constrained Optimal Allocation

Neyman allocation assumes that the cost of surveying a household is identical across all strata. In practice, travel to remote mountain districts can cost several times more than urban interviews.

If we define c_h as the unit cost of interviewing a household in stratum h , we can use the cost-constrained optimal allocation formula to maximize precision for a fixed total budget:

$$n_h = n \times \frac{N_h S_h / \sqrt{c_h}}{\sum_{h=1}^H N_h S_h / \sqrt{c_h}} \quad (9)$$

Under this design, strata with higher interview costs (c_h) receive relatively smaller sample sizes, allowing resources to go further in more cost-effective areas.

Suppose the unit costs of interviewing in our five strata are:

- Stratum 1 (Mountain Rural): $c_1 = \text{NPR } 8,000$
- Stratum 2 (Hill Rural): $c_2 = \text{NPR } 4,000$
- Stratum 3 (Hill Urban): $c_3 = \text{NPR } 3,000$
- Stratum 4 (Terai Rural): $c_4 = \text{NPR } 2,500$
- Stratum 5 (Terai Urban): $c_5 = \text{NPR } 2,000$

Table 12: Cost-Constrained Optimal Allocation Workflow

Stratum	Description	Product ($N_h S_h$)	Unit Cost (c_h)	Factor $N_h S_h / \sqrt{c_h}$	Final n_h
1	Mountain Rural	960,000	8,000	10,733	17
2	Hill Rural	5,100,000	4,000	80,664	129
3	Hill Urban	5,720,000	3,000	104,451	167
4	Terai Rural	14,040,000	2,500	280,800	448
5	Terai Urban	12,320,000	2,000	275,486	439
Total				752,134	1,200

Mountain Rural's sample size drops to just 17 households because it has low population variance ($S_h = 12.0$) and high interview costs ($c_1 = \text{NPR } 8,000$).

4.3 Minimum Sample Size Floors

While cost-constrained optimal allocation is highly efficient, a sample size of $n_1 = 17$ in Mountain Rural is too small to calculate stable standard errors or support sub-group analyses.

To address this, we define a **minimum sample floor** ($n_{\min}$) for all strata, typically ranging from 30 to 50 households depending on the survey type.

Table 13: Typical Minimum Sample Floors by Survey Type

Survey Type	Typical Floor ($n_{\min}$)
Household Living Standards Surveys (NLSS, NLFS)	30 – 50
Demographic and Health Surveys	40 – 60
Enterprise and Business Surveys	20 – 30
Agricultural Crop Yield Surveys	25 – 40

The Multi-Stage “Top-Up and Re-Allocate” Procedure: If any stratum falls below the minimum floor $n_{\min}$:

1. Set the sample size for those strata to exactly $n_{\min}$.
2. Subtract these fixed allocations from the total sample size to calculate the remaining sample budget:

$$n_{\text{remaining}} = n - \sum n_{\min} \quad (10)$$

3. Re-allocate the remaining sample budget only among the unconstrained strata using the original allocation formula.
4. Check if any of the newly allocated sample sizes fall below $n_{\min}$, and repeat the process if necessary.

Using our cost-constrained results and setting $n_{\min} = 30$:

- Mountain Rural is below the floor ($17 < 30$). We set its sample size to $n_1 = 30$.
- The remaining sample size to allocate is: $1,200 - 30 = 1,170$.
- We re-allocate the 1,170 units across Strata 2 to 5 based on their original allocation ratios.

The sample size for Mountain Rural is successfully increased to 30 households to support regional estimation, with the small difference distributed proportionally across the remaining strata.

Table 14: Floor-Adjusted Allocation Results ($n_{\min} = 30$)

Stratum	Description	Original Allocation	Adjusted Allocation	Net Change
1	Mountain Rural	17	30	+13
2	Hill Rural	129	127	-2
3	Hill Urban	167	165	-2
4	Terai Rural	448	443	-5
5	Terai Urban	439	435	-4
Total		1,200	1,200	

5 Implementing Allocation in Microsoft Excel

This section provides step-by-step instructions for building a dynamic, formula-driven sample allocation spreadsheet in Microsoft Excel. This allows users to change parameters like total sample size (n) or power (q) and see the calculations update automatically.

5.1 Workbook Architecture and Layout

We begin by setting up the sheet parameters. Place key global variables in columns Q and R to keep them separate from the main data tables.

Table 15: Excel Global Parameter Blocks

Cell Coordinate	Label (Col Q)	Assigned Value (Col R)
R1	Total Sample Size (n)	1200
R2	Power Parameter (q)	0.5
R3	Minimum Floor ($n_{\min}$)	30

Next, construct the column headers in Row 1 (Columns A to P):

- A: Stratum ID
- B: Description
- C: Population (N_h)
- D: Standard Deviation (S_h)
- E: Unit Cost (c_h)
- F: Equal allocation n_h
- G: Population share W_h
- H: Proportional allocation n_h
- I: Product $N_h S_h$
- J: Neyman proportion
- K: Neyman allocation n_h

- L: Square root population $\sqrt{N_h}$
- M: Power proportion
- N: Power allocation n_h
- O: Cost-constrained product $N_h S_h / \sqrt{c_h}$
- P: Cost-constrained allocation n_h

Populate rows 2 to 6 with the stratum information from Section 2.1, and reserve Row 7 for column totals.

5.2 Excel Formulas (Row 2 Examples)

To ensure the workbook updates dynamically, enter the following formulas in Row 2 and copy them down through Row 6:

- **Equal Allocation (Col F):**

=ROUND(\$R\$1 / COUNT(\$C\$2:\$C\$6), 0)

- **Population Share (Col G):**

=C2 / SUM(\$C\$2:\$C\$6)

- **Proportional Allocation (Col H):**

=ROUND(\$R\$1 * G2, 0)

- **Neyman Product (Col I):**

=C2 * D2

- **Neyman Proportion (Col J):**

=I2 / SUM(\$I\$2:\$I\$6)

- **Neyman Allocation (Col K):**

=ROUND(\$R\$1 * J2, 0)

- **Power Term N_h^q (Col L):**

=C2^\$R\$2

- **Power Proportion (Col M):**

=L2 / SUM(\$L\$2:\$L\$6)

- **Power Allocation (Col N):**

=ROUND(\$R\$1 * M2, 0)

- **Cost-Constrained Term (Col O):**

=(C2 * D2) / SQRT(E2)

- **Cost-Constrained Allocation (Col P):**

=ROUND(\$R\$1 * (O2 / SUM(\$O\$2:\$O\$6)), 0)

5.3 Formulas for Column Totals (Row 7)

In Row 7, enter the following sum formulas to verify the allocations sum to the target sample size:

- Population check: =SUM(C2:C6) (should equal 1,900,000)
- Allocations checks: =SUM(F2:F6), =SUM(H2:H6), =SUM(K2:K6), =SUM(N2:N6), =SUM(P2:P6) (should all equal 1,200)

5.4 Spreadsheet Design Tips for Training

- **Dynamic Demonstrations:** In cell R2, change q from 0.5 to 1.0 to show participants how the power allocation column dynamically updates to match the proportional allocation values. Changing q to 0 will show it update to match equal allocation.
- **Formulas for Design Weights:** Build helper columns for sampling fractions (=H2/C2) and design weights (=C2/H2) to show how the different allocation methods affect the final weights.
- **Protecting Formulas:** Lock the formula cells and unlock only the input cells (C2:E6 and R1:R3) before sharing the workbook. This allows users to experiment with parameters without accidentally breaking the calculations.

6 Implementing Allocation in Stata

While Excel is excellent for interactive learning, Stata is much more powerful for automating iterative loops (like minimum sample flooring), drawing samples from a frame, and setting up survey design parameters using `svyset`.

This section provides a complete, annotated Stata script to run the entire allocation and selection process.

```
1  * =====
2  * NEPAL HOUSEHOLD INCOME SURVEY: SAMPLE ALLOCATION & SELECTION WORKFLOW
3  * Author: Dr. Anil Shrestha
4  * =====
5  clear all
6  set more off
7  set seed 20830310          // Set seed for reproducibility
8
9  * -----
10 * STEP 1: Enter Stratum-Level Baseline Parameters
11 * -----
12 input str20 description double N_h double S_h double c_h
13     "Mountain Rural"      80000    12    8000
14     "Hill Rural"          340000    15    4000
15     "Hill Urban"          260000    22    3000
16     "Terai Rural"         780000    18    2500
17     "Terai Urban"         440000    28    2000
18 end
19
20 gen stratum = _n
21
22 * Define Global Sample Parameters
23 global n_total = 1200        // Target sample size
24 global q_power = 0.5         // Power exponent for power allocation
25 global n_min    = 30         // Minimum sample size floor
26
27 * -----
28 * STEP 2: Equal and Proportional Allocations
29 * -----
30 quietly count
31 local H = r(N)
32 gen n_equal = round($n_total / `H', 1)
33
34 quietly summarize N_h
35 local N_total = r(sum)
36 gen W_h = N_h / `N_total'
37 gen n_prop = round($n_total * W_h, 1)
38
39 * -----
40 * STEP 3: Neyman and Power Allocations
41 * -----
42 gen NhSh = N_h * S_h
43 quietly summarize NhSh
44 local sum_NhSh = r(sum)
45 gen n_neyman = round($n_total * (NhSh / `sum_NhSh'), 1)
46
47 gen Nh_q = N_h ^ $q_power
48 quietly summarize Nh_q
49 local sum_Nhq = r(sum)
50 gen n_power = round($n_total * (Nh_q / `sum_Nhq'), 1)
```

```

51
52 * -----
53 * STEP 4: Cost-Constrained Allocation
54 * -----
55 gen NhSh_cost = N_h * S_h / sqrt(c_h)
56 quietly summarize NhSh_cost
57 local sum_NhSh_cost = r(sum)
58 gen n_cost = round($n_total * (NhSh_cost / `sum_NhSh_cost'), 1)
59
60 * -----
61 * STEP 5: Iterative Minimum Floor Loop (Applied to Neyman)
62 * -----
63 gen n_floor = $n_total * (NhSh / `sum_NhSh') // Start with raw Neyman
64 local converged = 0
65 local iter = 0
66
67 while `converged' == 0 {
68     local iter = `iter' + 1
69     gen byte is_floored = (n_floor < $n_min)
70
71     quietly summarize is_floored
72     local num_floored = r(sum)
73     local allocated_to_floored = `num_floored' * $n_min
74     local n_remaining = $n_total - `allocated_to_floored'
75
76     quietly summarize NhSh if is_floored == 0
77     local sum_NhSh_free = r(sum)
78
79     replace n_floor = $n_min if is_floored == 1
80     replace n_floor = `n_remaining' * (NhSh / `sum_NhSh_free') if
is_floored == 0
81
82     drop is_floored
83     quietly count if n_floor < $n_min
84     if r(N) == 0 {
85         local converged = 1
86     }
87 }
88 gen n_neyman_floor = round(n_floor, 1)
89
90 * Save allocations for the selection step
91 preserve
92 keep stratum n_neyman_floor
93 rename n_neyman_floor n_allocated
94 save "allocation_results.dta", replace
95 restore
96
97 * -----
98 * STEP 6: Simulate a Frame and Draw the Sample
99 * -----
100 clear
101 * Simulate a population frame of 1,900 units for this demonstration
102 set obs 1900
103 gen hh_id = _n
104 gen stratum = 1 + (hh_id > 80) + (hh_id > 420) + (hh_id > 680) + (hh_id >
1460)
105
106 * Merge the final allocations into the frame

```

```

107 merge m:1 stratum using "allocation_results.dta", nogenerate
108
109 * Run Stratified Simple Random Sampling
110 gen rand_draw = runiform()
111 bysort stratum (rand_draw): gen selection_rank = _n
112 gen hh_selected = (selection_rank <= n_allocated)
113
114 * -----
115 * STEP 7: Define Weights and Declare Survey Design
116 * -----
117 * Set the population-to-sample scaling factor (1:1000 for this simulation)
118 gen N_h_real = 80000 if stratum == 1
119 replace N_h_real = 340000 if stratum == 2
120 replace N_h_real = 260000 if stratum == 3
121 replace N_h_real = 780000 if stratum == 4
122 replace N_h_real = 440000 if stratum == 5
123
124 gen weight = N_h_real / n_allocated
125
126 * Set up the complex survey design
127 svyset hh_id [pweight=weight], strata(stratum)
128 svydescribe

```

Complete Stratified Sample Allocation Script (Stata)

7 Edge Cases, Practical Issues, and Quality Checks

In the field, sample designs are rarely as clean as theoretical calculations. This section covers common practical challenges and how to address them.

7.1 Rounding Drift and the Largest Remainder Method

Because sample sizes must be whole numbers, rounding individual strata allocations (n_h) to integers can cause the sum to drift away from the target total (n). This can leave the sample size slightly over or under budget.

To fix this systematically, researchers use the **Largest Remainder Method**:

1. Round all raw stratum sample sizes (n_h) **down** to the nearest integer.
2. Sum these rounded values and calculate the shortfall:

$$\text{Shortfall} = n - \sum \lfloor n_h \rfloor \quad (11)$$

3. Calculate the fractional remainder for each stratum: $n_h - \lfloor n_h \rfloor$.
4. Sort the strata by their fractional remainders in descending order.
5. Add one unit to each of the top strata until the shortfall is fully resolved.

Table 16: Shortfall Adjustment (Largest Remainder Method)

Stratum	Description	Raw n_h	Rounded Down	Remainder	Final n_h
1	Mountain Rural	30.20	30	0.20	30
2	Hill Rural	160.46	160	0.46	160
3	Hill Urban	180.00	180	0.00	180
4	Terai Rural	441.71	441	0.71	441 + 1 = 442
5	Terai Urban	387.60	387	0.60	387 + 1 = 388
Total		1,200.00	1,198	(Shortfall = 2)	1,200

Under proportional allocation, the rounded-down sum is 1,198, leaving a shortfall of 2 units.

7.2 Over-Allocation ($n_h > N_h$) and "Take-All" Strata

For small strata with high variability, optimal allocation formulas can calculate a sample size larger than the stratum's actual population ($n_h > N_h$).

When this occurs, we apply the **Take-All** adjustment:

1. Designate the over-allocated stratum as a "Take-All" stratum, sampling every unit in its population ($n_h = N_h$).
2. Convert the stratum into a census segment, removing it from the sampling design.
3. Subtract its population from the total sample budget: $n_{\text{remaining}} = n - N_h$.
4. Re-allocate the remaining sample budget among the remaining strata.

This adjustment can be automated in Stata using a cap-and-redistribute loop, similar to the minimum floor loop in Section 6.

7.3 Zero- or Near-Zero Allocation

Strata with very small populations or extremely low variance can receive an allocated sample size of $n_h = 0$ or 1 under optimal formulas.

This is a serious design error:

- If $n_h = 0$, the stratum is completely unrepresented, which biases national estimates.
- If $n_h = 1$, we can calculate a point estimate but **cannot estimate standard errors or variance** for that stratum, as there is no variation within a single unit.

Applying a minimum sample floor (Section 4.3) ensures that no stratum receives a sample size below $n_{\min} \geq 30$, preventing this issue.

7.4 Final Quality-Assurance Verification Checks

Before sending an allocation plan to field teams, verify that it passes the checks in Table 17.

Table 17: Required Pre-Fieldwork Quality Checks

Check Area	Verification Target and Action
Sample Budget Alignment	The sum of the allocated sample sizes must equal the total budget exactly: $\sum n_h = n$.
Design Weight Validity	Verify that no stratum has $n_h = 0$, which would cause division errors (#DIV/0!) when calculating weights.
Under-Allocation Check	Ensure every stratum meets the minimum floor requirements: $n_h \geq n_{\min}$.
Over-Allocation Check	Ensure no stratum has a sample size larger than its population: $n_h \leq N_h$.
Cost Constraints	For cost-optimal designs, confirm that the projected fieldwork cost ($\sum n_h \times c_h$) is within the approved budget.
Weight Dispersion (DEFF)	Check that the ratio of the maximum design weight to the minimum design weight is reasonable (ideally < 4.0). High weight variation can drastically inflate standard errors.

Estimating Weight-Based Design Effects: You can approximate the design effect (DEFF) caused by weight variation using Kish's approximation formula:

$$\text{DEFF}_{\text{weights}} \approx 1 + \text{CV}(w_h)^2 \quad (12)$$

where $\text{CV}(w_h)$ is the coefficient of variation of the design weights across the selected sample. A DEFF value above 2.0 indicates that the allocation design introduces high variance, and may need to be adjusted back toward a proportional design.

8 Pre-Fieldwork Design Checklist

Use this practical checklist to verify your sample allocation design before starting fieldwork:

- ☐ **Total Sample Matching:** The sum of individual stratum allocations ($\sum n_h$) matches the contractually approved national sample size n exactly.
- ☐ **Minimum Floors Met:** Every stratum meets or exceeds the minimum sample size required for stable variance calculation ($n_h \geq n_{\min}$).
- ☐ **Zero-Selection Check:** No stratum has been assigned 0 or 1 unit.
- ☐ **Over-Allocation Resolved:** No stratum has $n_h > N_h$. All census-type segments have been set to $n_h = N_h$ and removed from the active sampling design.
- ☐ **Weight Calculations Complete:** Initial design weights ($w_h = N_h/n_h$) have been calculated for all strata and are positive, finite numbers.
- ☐ **Budget Feasibility:** The total projected cost of the allocation design ($\sum n_h \times c_h$) is within the approved budget.
- ☐ **Design Declaration Setup:** The survey design parameters, including strata, clusters, and weights, have been correctly configured in the statistical software (e.g., using `svyset` in Stata).
- ☐ **Document Control:** The final allocation spreadsheet has been locked, and the Stata scripts are version-controlled to prevent accidental changes.